

Lab 02 — Tags and Resource Locks

Hands-on Azure lab covering resource tags and resource locks.

Objective

Practice organizing Azure resources with tags, and protecting resources from accidental changes or deletion using resource locks.

Azure resources used

Resource	Name
Resource group	rg-gp-tags-locks
Storage account (first)	stgptagslock65624538
Storage account (second)	stgptagsops65624538

Tags

Resource	department	environment
rg-gp-tags-locks	development	test
stgptagslock65624538	development	test
stgptagsops65624538	operations	test

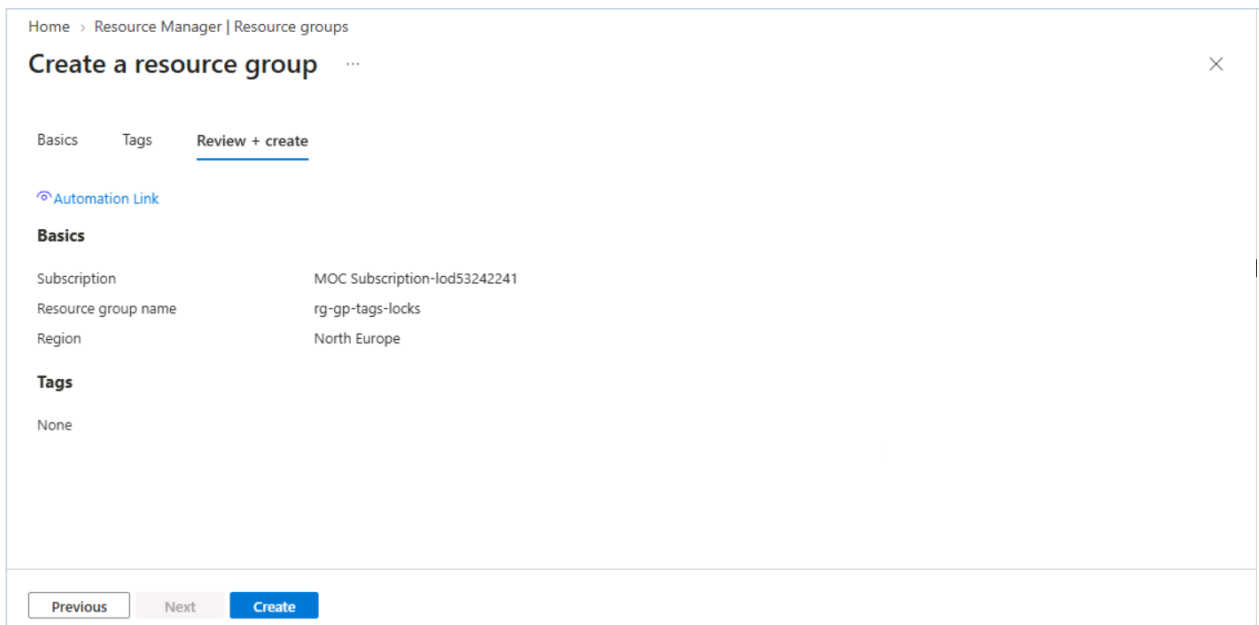
Locks

Lock name	Lock type	Applied to
prevent-delete	Delete	Storage account stgptagslock65624538
read-only-rg	Read-only	Resource group rg-gp-tags-locks

This was a hands-on lab in a Microsoft Azure training environment, following guided steps. It was not a production setup.

1. Creating the Resources

I created the resource group **rg-gp-tags-locks** in North Europe, then created two Azure storage accounts inside it.



Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

[Automation Link](#)

Basics

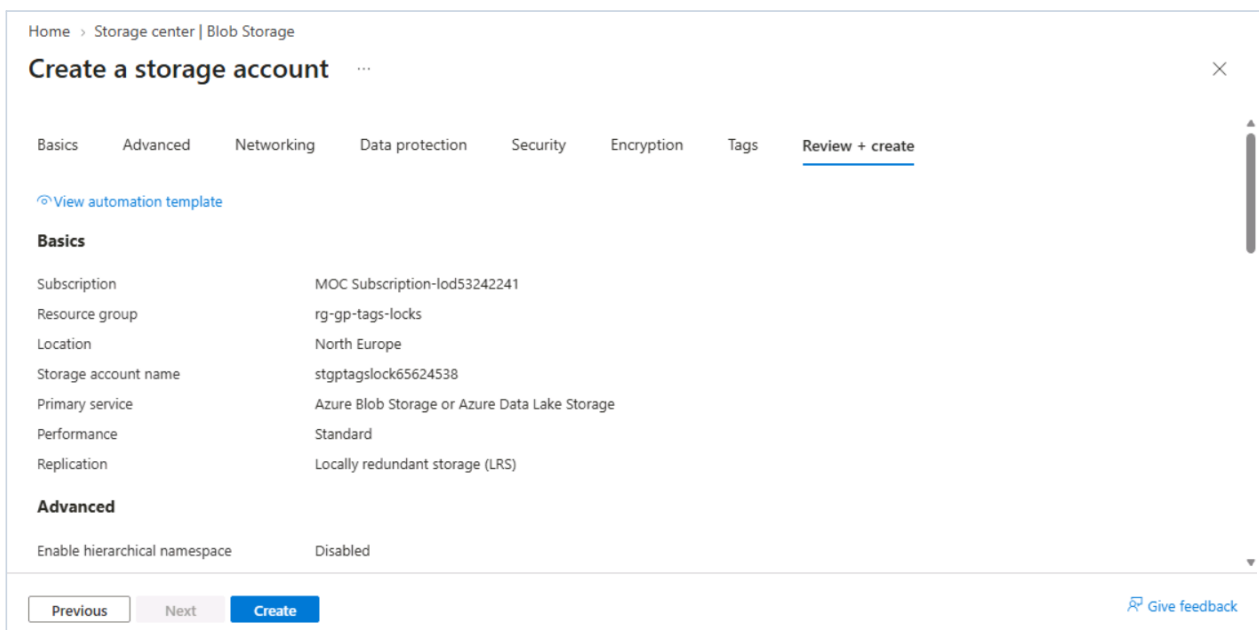
Subscription	MOC Subscription-lod53242241
Resource group name	rg-gp-tags-locks
Region	North Europe

Tags

None

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Creating the resource group rg-gp-tags-locks in North Europe.



Home > Storage center | Blob Storage

Create a storage account

Basics Advanced Networking Data protection Security Encryption Tags Review + create

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Basics

Subscription	MOC Subscription-lod53242241
Resource group	rg-gp-tags-locks
Location	North Europe
Storage account name	stgptagslock65624538
Primary service	Azure Blob Storage or Azure Data Lake Storage
Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace	Disabled
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Creating the first storage account, stgptagslock65624538: Standard performance, LRS.

2. Tags

Tags are name/value pairs that help organize resources. I added **department** and **environment** tags to the resource group and to both storage accounts.

The screenshot shows the 'Tags' page for the resource group 'rg-gp-tags-locks'. The left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags (selected), Analyze (Preview), Resource visualizer, Events, Settings, Cost Management, Monitoring, Automation, and Help. The main content area has a search bar, 'Delete all', and 'Feedback' buttons. Below these, there is explanatory text about tags and a warning. A table with two columns, 'Name' and 'Value', contains the following entries:

Name	Value
department	development
environment	test

Below the table, the resource group 'rg-gp-tags-locks' is listed with its tags: 'department : development' and 'environment : test'. At the bottom are 'Apply' and 'Discard changes' buttons.

Resource group tags: department = development, environment = test.

The screenshot shows the 'Tags' page for the storage account 'stgptagsops65624538'. The left sidebar contains navigation links: Overview, Activity log, Tags (selected), Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Storage Mover, Partner solutions, Resource visualizer, Data storage, and Security + networking. The main content area has a search bar, 'Delete all', and 'Feedback' buttons. Below these, there is explanatory text about tags and a warning. A table with two columns, 'Name' and 'Value', contains the following entries:

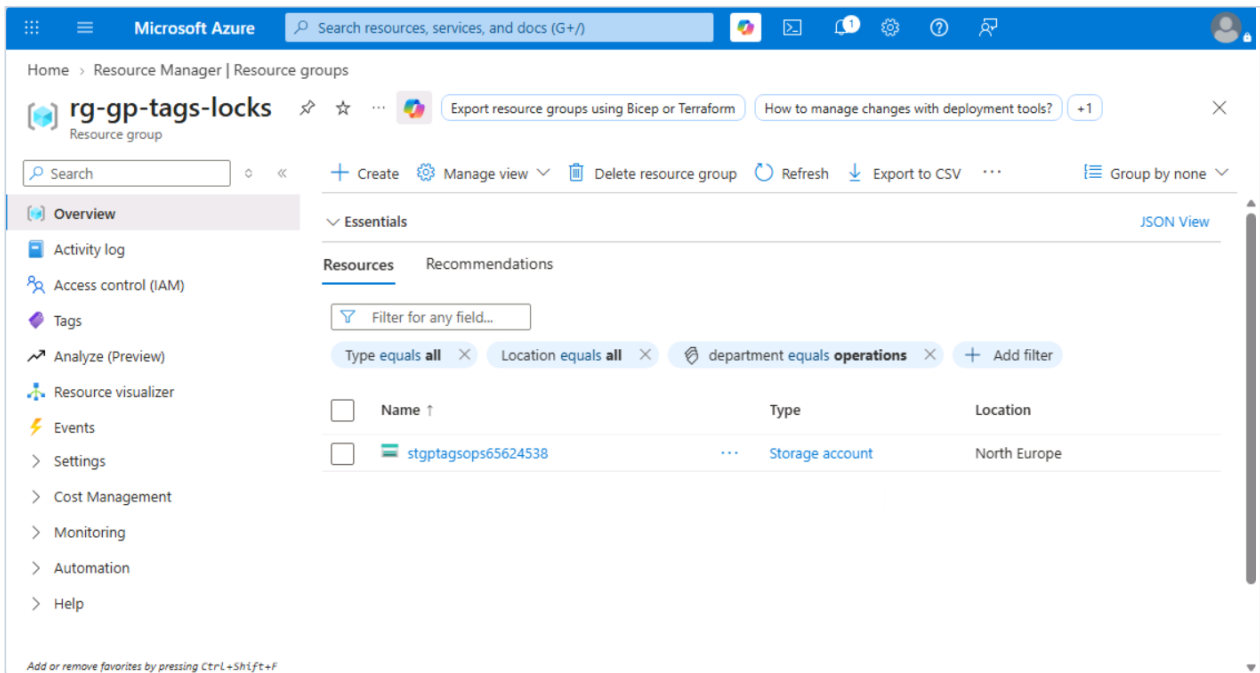
Name	Value
department	operations
environment	test

Below the table, the storage account 'stgptagsops65624538' is listed with its tags: 'department : operations' and 'environment : test'. At the bottom are 'Apply' and 'Discard changes' buttons.

Second storage account tags: department = operations, environment = test.

3. Filtering Resources by Tag

With the tags in place, I filtered the resources in the resource group by **department = operations**. Only the second storage account, **stgptagsops65624538**, was listed.



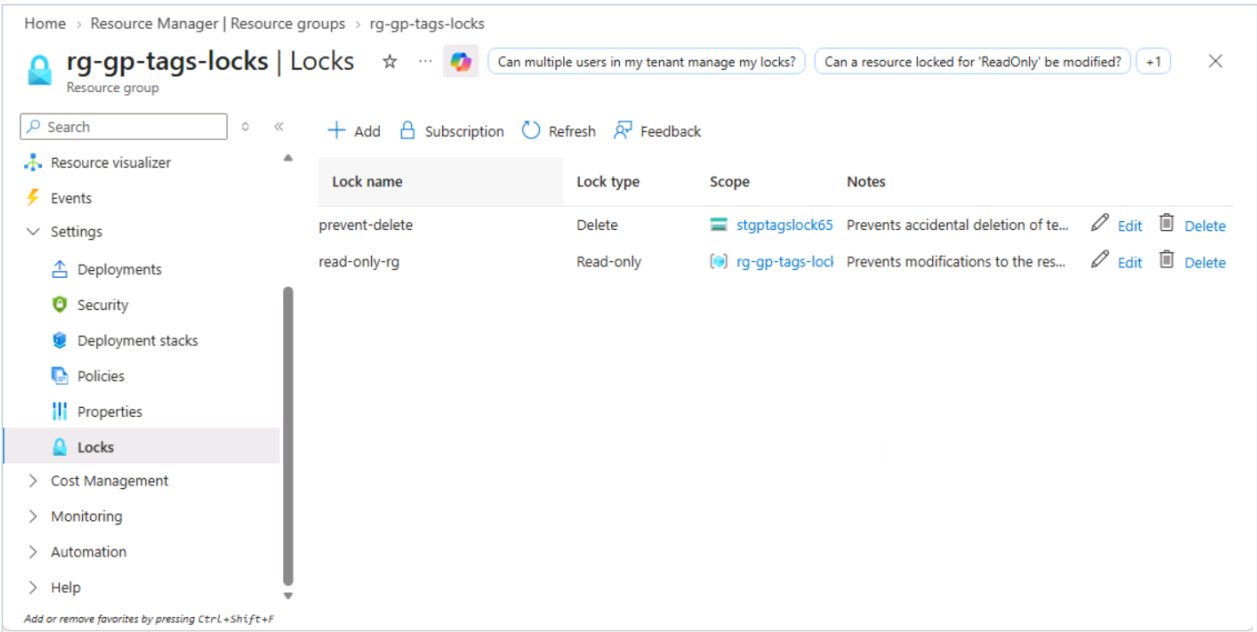
The screenshot shows the Microsoft Azure portal interface for the resource group **rg-gp-tags-locks**. The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, and more. The main area displays the 'Essentials' section with a 'Resources' tab selected. A filter is applied: **department equals operations**. Below the filter, a table lists the resources. Only one resource is visible: **stgptagsops65624538**, which is a **Storage account** located in **North Europe**.

Name ↑	Type	Location
stgptagsops65624538	Storage account	North Europe

Resource group overview filtered by department = operations.

4. Resource Locks

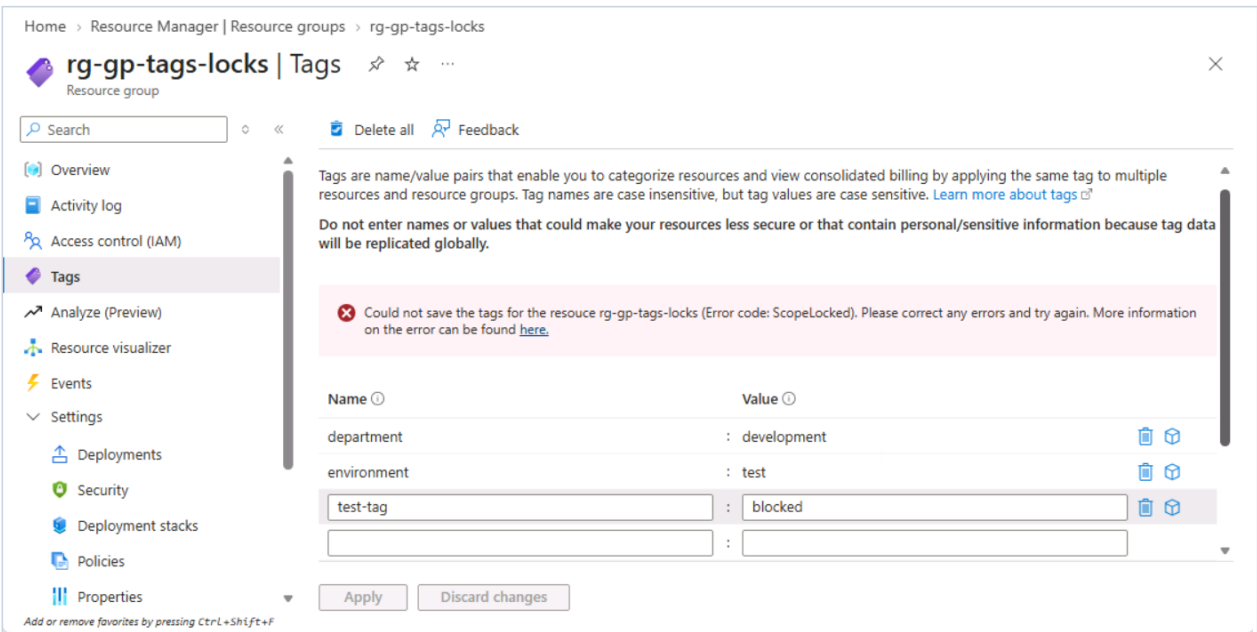
Locks protect resources from accidental changes. I created a **Delete** lock named **prevent-delete** on the storage account, and a **Read-only** lock named **read-only-rg** on the resource group.



Both locks listed: prevent-delete (Delete) and read-only-rg (Read-only).

Testing the read-only lock

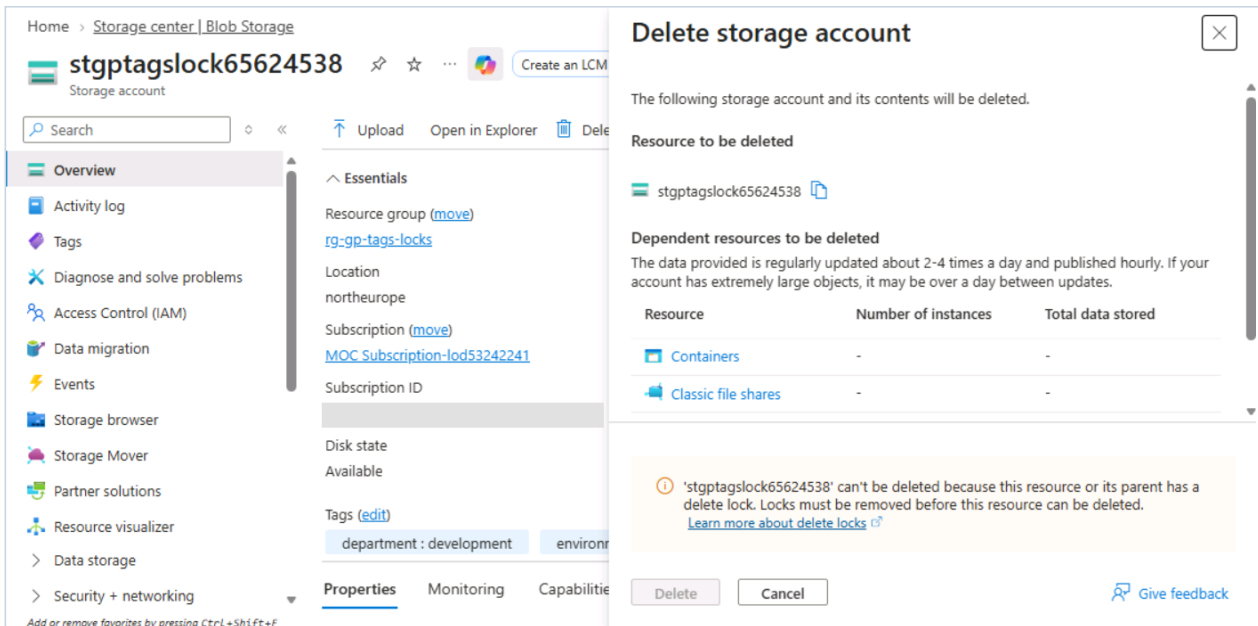
I tried to change the resource group by adding a new tag (test-tag = blocked). Azure would not save it and returned the error code **ScopeLocked**.



Adding a tag to the resource group failed with the error code ScopeLocked.

5. Testing the Delete Lock

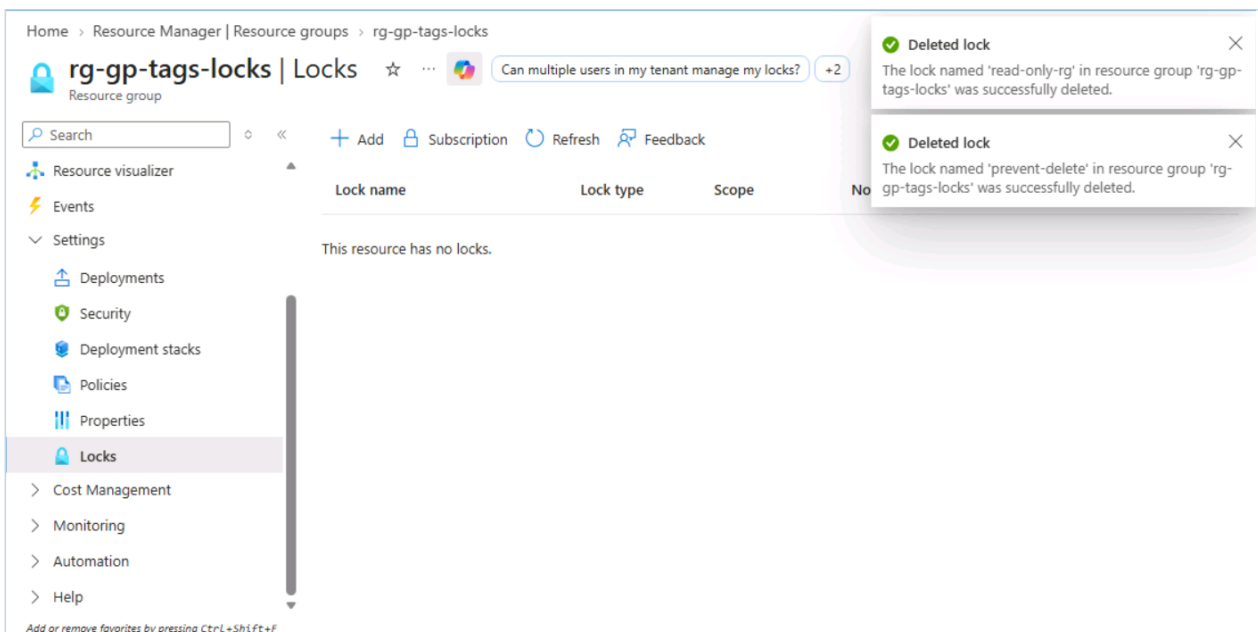
I tried to delete the protected storage account **stgptagslock65624538**. The **Delete** button stayed disabled, and a message said the delete lock must be removed first.



Delete storage account panel: deletion blocked by the delete lock.

Removing the locks

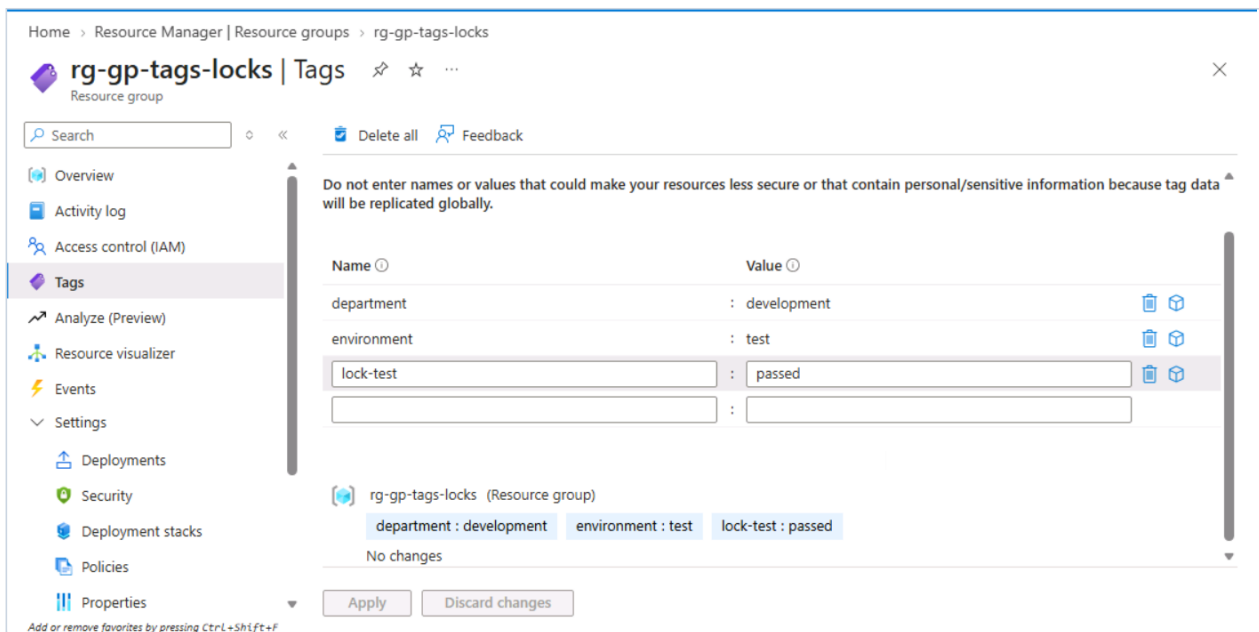
After the tests, I removed both locks. The Locks page then showed no locks, and Azure confirmed that read-only-rg and prevent-delete were deleted.



Both locks deleted; the resource group now has no locks.

6. Verifying Access After Removing the Locks

To check that normal changes worked again, I added a new tag to the resource group (lock-test = passed). This time there was no error.



Adding the tag lock-test = passed worked once the locks were removed.

What I Learned

- How to use Azure tags to label resources and filter them.
- How a Delete lock stops a resource from being deleted.
- How a Read-only lock stops changes, such as editing tags.
- How to test protection by trying the blocked action and reading the error.
- How to remove locks and confirm that normal changes work again.

Conclusion

This lab gave me practice with tags and locks in a training environment. I could see the locks block a change and a deletion, and see normal access return once they were removed.